A Finite-Horizon, Cost-Aware Separation Technical Appendix to Machine Markets

Research development draft for author and professor review

29 October 2026

Scope. An analytic result under an explicit controlled-measurement and inventory-service contract. Novelty and independent proof review remain open. This document does not claim a foundational contribution or real-market validation.

Finite-horizon information, cohort ambiguity, and net inventory-service value

Status and contribution boundary. This appendix supplies a restricted analytic separation and its financial consequence. The proof below is written in full; exact rational and numerical checks accompany it. It has not received independent proof review and is not a Lean formalization. Its priority is unconfirmed. Control without unique identification, task-directed exploration, Gaussian functional estimation, moment matching, and the value of information are established ideas (van Waarde et al., 2020; Wagenmaker et al., 2021; Donoho, 1994; Heinrich and Kahn, 2018; Blackwell, 1953). The candidate extension is their explicit combination for a finite-horizon inventory commitment, with a hard stress budget and the information bill deducted from both capital and profit. It does not close the broader open theorem of the preceding drafts.

The observation contract and the financial institution

Fix $H \geq 4$, $0 \leq a < 1$, $\kappa > 0$ and $0 \leq Q \leq \overline{Q} < (1-a)/\kappa$. There are arbitrarily many finite cohorts, with $q_i \geq 0$, $\sum_i q_i = Q$, and $0 < \eta_i \leq 1$:

$$x_{t+1} = ax_t + \kappa\big(\sum_i q_i y_{i,t} + u_t\big), \qquad y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}.$$

Initial states are reset to zero. Let $h_0, \ldots, h_{H-1}$ be the price response to a unit impulse, with $h_0 = \kappa$, and put $G = \sum_{t<H} h_t$. Positivity implies $h_t \geq 0$. No positive lower bound on η_i is imposed.

The observer purchases N independent reset experiments, each lasting H model periods and returning all H taps with independent Gaussian errors of known variance σ^2. Thus the tap means are $\overline{h}_t = h_t + \sigma Z_t/\sqrt{N}$, with independent standard normal Z_t. The same observations are available for identifying the cohorts and for choosing the financial position. One experiment is not one scalar reading: it returns H readings and costs c_H, including the declared operating and time charges. Total information cost is $F = Nc_H$. This observation contract is not an ordinary exchange tape; longer horizons cost both time and readings. Plant parameters, the feedback cap, reset, and stationarity are assumptions. Measurements are acquired ahead of a common future service-start

deadline. Cash can earn the common reference rate during acquisition; the model does not include earlier service opportunities. The stated measurement price is exogenous, not a calibrated self-impact cost for actual market probes.

The institution is a price-taking, unlevered inventory-warehousing desk. It buys $I \geq 0$ units at P_0, holds them for H periods, and closes. It earns a prequoted service fee pI *after* settlement. Round-trip execution and carry costs are $bI + dI^2$, where $b \geq 0$ and $d > 0$. For example, $b = b_{\text{trade}} + P_0 r_f H/(365 \cdot 24 \cdot 60)$ for one-minute model periods. The service premium is unavailable as collateral until after settlement. Both trading costs and F must be prefunded. Initial cash is W, the stressed-settlement loss budget is B, and $|u_t| \leq \overline{u}$, with $\overline{u} > 0$. Assume $P_0 > \overline{u}\,\overline{G}$, so the asset price remains positive. This desk does not alter market feedback, depth, or another participant's behaviour. The service fee is exogenous, not an equilibrium price.

For the expected-profit objective, future shocks have conditional mean zero given the observations. Consequently the conditional expected trading price change is zero. The financial objective, including every declared charge, is

$$V(I, F) = g(I) - F, \qquad g(I) = (p - b)I - dI^2.$$

This is expected service profit with a hard stress constraint, not a Kelly growth theorem or a return-prediction result. There are no additional training or validation tapes, no leverage, uncertain fills, or changing inventory.

Lemma 1 (Sharp prior bounds and exact prefunded capacity)**.** *Write*

$$\underline{G} = \kappa \sum_{t<H} a^t, \qquad \overline{G} = \kappa \sum_{t<H} (a + \kappa\overline{Q})^t.$$

Then $\underline{G} \leq G \leq \overline{G}$, and both endpoints are attained in the declared class. For a gain bound $C \geq G$ and $0 \leq F \leq \min(B, W)$, define

$$R(v, r) = \frac{2v}{r + \sqrt{r^2 + 4dv}}, \qquad I_u = \frac{(p - b)_+}{2d},$$
$$I(C, F) = \min\{I_u, R(B - F, \overline{u}C + b), R(W - F, P_0 + b)\}.$$

This is the optimal expected-profit position subject to the stated stress reserve and prefunding constraints. The zero-violation no-observation robust optimum is $I_{\text{nd}} = I(\overline{G}, 0)$, not an arbitrarily conservative benchmark. The response-informed oracle is $I_\circ = I(G, 0)$; it knows no future shocks.

Proof. Let $k_j = \sum_i q_i \eta_i (1 - \eta_i)^j$. The normalized price response is the renewal sequence with defective interarrival probabilities $s_1 = a + \kappa k_0$ and $s_j = \kappa k_{j-1}$ for $j \geq 2$. Their total mass is $a + \kappa Q < 1$, with the remaining mass placed at infinity. The cumulative interarrival distribution at integer $j \geq 1$ is

$$a + \kappa \sum_i q_i [1 - (1 - \eta_i)^j].$$

It lies between a and $a + \kappa\overline{Q}$. Coupling interarrivals by the same independent uniforms orders every renewal time and hence the number of renewals before H. Taking expectations proves the two bounds. The lower endpoint is the no-feedback economy; the upper is one cohort with $Q = \overline{Q}$ and $\eta = 1$. This also proves sharpness of the no-observation stress bound.

At date $t \leq H$, the price change is $x_t = \sum_{j<t} h_j u_{t-1-j}$, so $|x_t| \leq \overline{u} \sum_{j<t} h_j \leq \overline{u}G$. All costs paid by that date are at most the total prefunded costs. The constant adverse sequence $u_t = -\overline{u}$

attains the total price loss $I\overline{u}G$ at settlement. Before the service fee is credited, worst loss from initial wealth is therefore exactly $I\overline{u}G + bI + dI^2 + F$. The declared robust reserve requirement is $I\overline{u}C + bI + dI^2 + F \le B$; prefunding requires $P_0 I + bI + dI^2 + F \le W$. These increasing quadratics yield the two roots. Finally g increases up to I_u and decreases after it, proving the optimizer and the two benchmarks. This exact reserve contract is not a peak-to-trough drawdown or a repeated-deployment guarantee. □

Lemma 2 (No data at the same error budget)**.** *Let $I_f = \min\{I_u, R(W, P_0 + b)\}$ and allow a data-free rule to randomize, with probability of a stress-budget violation at most γ in every economy of the class. Its greatest expected service profit is*

$$V_{\mathrm{nd},\gamma} = (1-\gamma)g(I_{\mathrm{nd}}) + \gamma g(I_f).$$

It is attained by the displayed two-position lottery. This is a theoretical benchmark, not a recommended trading policy. The corresponding response- informed same-error-budget oracle has value $(1-\gamma)g(I_\circ) + \gamma g(I_f)$.

Proof. Truncating an action at I_f weakly improves profit and lowers risk, so restrict to $0 \le I \le I_f$. At the attainable worst economy with gain $\overline{G}$, every $I > I_{\mathrm{nd}}$ is unsafe unless $I_{\mathrm{nd}} = I_f$, in which case all larger actions are already dominated. Thus a feasible lottery puts at most γ mass above I_{nd}. Below that point profit is at most $g(I_{\mathrm{nd}})$, and above it at most $g(I_f)$. This gives the upper bound. The two-point lottery attains it and is unsafe in any economy with probability at most γ. Replace $\overline{G}$ by the known G for the oracle statement. □

Theorem 1 (Uniform cost-aware finite-horizon decision)**.** *Fix $0 < \gamma < 1/2$, $z = \Phi^{-1}(1-\gamma)$, $s_N = \sigma\sqrt{H/N}$, and*

$$Y = \sum_{t<H} \overline{h}_t, \qquad C_N = \min\{\overline{G}, \max\{\underline{G}, Y + z s_N\}\}.$$

Purchase a fixed, affordable N and take $I_N = I(C_N, F)$. The following hold uniformly over the declared positive-cohort class, without dependence on cohort spacing, cohort count, or a memory-rate floor.

1. *With probability at least $1-\gamma$ over measurements, the position satisfies the stress budget for* every *bounded future shock path.*

2. *Put $r_0 = \overline{u}\underline{G} + b$, $I_{\max} = W/P_0$,*

$$L_G = \frac{\overline{u} I_{\max}}{r_0}, \qquad L_F = \frac{1}{\min(r_0, P_0 + b)},$$
$$A = 2z\sigma\sqrt{H}\,(p-b)_+ L_G, \quad D = 1 + (p-b)_+ L_F.$$

   With probability at least $1-2\gamma$,

$$0 \le g(I_\circ) - V(I_N, F) \le A N^{-1/2} + D c_H N.$$

   For sufficiently small $c_H > 0$, choosing $N = \lceil (A/(2Dc_H))^{2/3} \rceil$ is affordable and gives a bound $3\,2^{-2/3} A^{2/3} (Dc_H)^{1/3} + Dc_H$, when $A > 0$. This is an upper bound from this procedure, not a new minimax optimality claim.

3. *Let $I_- = I(G + 2zs_N, F)$. A sufficient condition for strictly positive expected net information value relative to the strongest no-data rule at the* same *error budget is*

$$\underbrace{(1-2\gamma)g(I_-) - F - V_{\mathrm{nd},\gamma}}_{\mathcal{L}_N} > 0.$$

Precisely, $\mathbb{E}_D[V(I_N, F)] - V_{\mathrm{nd},\gamma} \geq \mathcal{L}_N$. The conclusion can fail at high data costs or a low service premium.

Proof. $Y - G$ is Gaussian with variance $H\sigma^2/N$. On $\{Y - G \geq -zs_N\}$, which has probability $1 - \gamma$, $C_N \geq G$; clipping preserves this because G lies between the prior bounds. Lemma 1 proves simultaneous safety for every future bounded path on this single event, including the data-selected position. There is no tail beyond H to estimate, explaining the absence of a memory-floor assumption. This does not certify an infinite horizon.

On $E = \{|Y - G| \leq zs_N\}$, of probability $1 - 2\gamma$, $G \leq C_N \leq G + 2zs_N$. The position function is decreasing in each of C and F, so $I_- \leq I_N \leq I_\circ$ on E. To bound the position difference, implicitly differentiate the stress root: its derivatives are $-\overline{u}I/(\overline{u}C + b + 2dI)$ in C and $-1/(\overline{u}C + b + 2dI)$ in F. Clip this root at $I_{\max}$ without changing the final minimum, since the prefunding root is at most $I_{\max}$. The clipped root is L_G-Lipschitz in C and $1/r_0$-Lipschitz in F. The prefunding root is independent of C and $1/(P_0 + b)$-Lipschitz in F. Minimum with I_u preserves these common Lipschitz bounds. Hence

$$0 \leq I_\circ - I_N \leq 2L_G zs_N + L_F F.$$

On $[0, I_u]$, $0 \leq g' \leq (p - b)_+$. Multiplication and addition of the paid fee give the regret bound. Minimizing $AN^{-1/2} + Dc_H N$ over positive real N gives the stated root and value. Rounding up adds at most Dc_H; affordability follows as $c_H N = O(c_H^{1/3}) \to 0$. The case $A = 0$ has no positive-gain information task and does not need this optimization.

On E, $g(I_N) \geq g(I_-)$, while $g(I_N) \geq 0$ always because $0 \leq I_N \leq I_u$. Thus $\mathbb{E}_D g(I_N) \geq (1 - 2\gamma)g(I_-)$. Subtracting F and the same-error-budget no-data value proves the last assertion. Conditional mean-zero future flows justify the stated expected-profit interpretation; their distribution is not learned from extra tapes. □

Theorem 2 (A two-cohort identification obstruction under the same observations)**.** *Fix $Q > 0$, $a + \kappa Q < 1$, and $0 < \eta_0 < 1$. For sufficiently small $\delta > 0$, compare the two economies*

	normalized weights	*adaptation rates*
A_δ	$(1/2, 1/2)$	$(\eta_0 - \delta, \eta_0 + \delta)$
B_δ	$(1/5, 4/5)$	$(\eta_0 - 2\delta, \eta_0 + \delta/2)$.

Actual weights are Q times those displayed. For fixed $H \geq 4$, $D_H(\delta) = \|h^{\mathrm{B}}_{0:H-1} - h^{\mathrm{A}}_{0:H-1}\|_2 = \Theta(\delta^3)$. The exact minimax error for distinguishing these two known *alternatives from N reset experiments is*

$$e_N = \Phi\left(-\frac{\sqrt{N} D_H(\delta)}{2\sigma}\right).$$

Therefore error at most $\beta < 1/2$ requires $N \geq [2\sigma\Phi^{-1}(1-\beta)/D_H(\delta)]^2 = \Theta(\delta^{-6})$. In particular any recovery of the two normalized weights to sup-norm error less than 0.15 *with probability at least $1 - \beta$ in both worlds must satisfy this testing requirement. Both worlds contain exactly two distinct rates. Both have identical infinite summed response, but their finite-horizon gains are not asserted identical.*

Proof. The two rate distributions have equal mass, mean η_0, and variance δ^2. Their third centered moments are respectively 0 and $-3\delta^3/2$. Thus their kernels k_j agree at $j = 0, 1$ and $k_2^{\mathrm{B}} - k_2^{\mathrm{A}} = -3Q\delta^3/2$. The response recursion gives equal h_0, h_1, h_2 and

$$h_3^{\mathrm{B}} - h_3^{\mathrm{A}} = -3\kappa^2 Q\delta^3/2,$$

which is a lower bound for D_H in absolute value.

For an upper bound put $f_t(\eta) = \eta(1-\eta)^t$. Its third derivative is $3t(t-1)(1-\eta)^{t-2} - \eta t(t-1)(t-2)(1-\eta)^{t-3}$, with absent low-degree terms zero. The sum over $t \geq 0$ of its absolute value is at most $12/\eta^3$. For $\delta \leq \eta_0/4$, all intervening rates are at least $v = \eta_0/2$. Taylor's integral remainder in ℓ_1 is therefore at most $2|\eta - \eta_0|^3/v^3$. The matching first two moments cancel the quadratic Taylor terms. The sum of the two third absolute centered moments is $(1 + 17/10)\delta^3 = 27\delta^3/10$, giving

$$\|k^{\mathrm{B}} - k^{\mathrm{A}}\|_1 \leq \frac{27Q}{5v^3}\delta^3.$$

If $K(z) = \sum k_t z^t$ and $H(z) = \sum h_t z^t$, the recursion gives $H(z) = \kappa/[1 - az - \kappa z K(z)]$. Consequently $H_{\mathrm{B}} - H_{\mathrm{A}} = zH_{\mathrm{A}}H_{\mathrm{B}}(K_{\mathrm{B}} - K_{\mathrm{A}})$. The infinite response sums of both economies equal $S = \kappa/(1 - a - \kappa Q)$. Young's inequality yields $D_H \leq S^2\|k^{\mathrm{B}} - k^{\mathrm{A}}\|_1 = O(\delta^3)$. This proves the two-sided order with constants independent of δ.

For equal-covariance Gaussian laws, the likelihood-ratio test projects the sample mean onto the difference of the two known means, and thresholds at their midpoint. The projected normal law gives the displayed error in each world. Its equal-prior Bayes optimality and symmetry establish minimax optimality. A weight estimator accurate within 0.15 distinguishes the two disjoint normalized-weight balls, proving the identification lower bound. Finite-horizon gains need not be equal: only their difference is of cubic order. The test concerns latent cohort recovery, not identification of the input-output response vector, and does not cover alternative probe designs. □

Corollary 1 (A priced separation)**.** *Fix $c_H > 0$ and an affordable N for which $\mathcal{L}_N > 0$ at the limiting merged economy $\delta = 0$. For all sufficiently small positive δ, the financial rule of Theorem 1 has a strictly positive expected net information value, with its stated safety probability. Its observation requirement has no dependence on δ. At this same fixed information price, sufficiently small δ makes cohort identification at error β unaffordable within the same hard fee budget B: $c_H[2\sigma\Phi^{-1}(1-\beta)/D_H(\delta)]^2 > B$. Any procedure limited to that many affordable observations is covered, including one that stops early, since it can be viewed as a function of the full affordable sample. A rational agent may abandon identification and use the no-data rule. There is no assertion that regulation requires identification.*

Proof. Apply the two theorems. The first statement uses the strictly positive lower bound on net value, rather than an assumption that more information is automatically beneficial. The second follows from $D_H = \Theta(\delta^3)$. For a neighbourhood of a fixed δ where the strict value inequality holds, continuity preserves it. To claim the inequality for all sufficiently small δ, it suffices that it hold strictly at the limiting merged economy: finite G_H and I_- are continuous there. The numerical example reports both the primary pair and that limiting check. No finite-class grid is used in the uniform decision guarantee. □

What the result does not establish. The direct gain-sum certificate is exactly ordinary Gaussian projection onto the vector of ones. It is not a new estimator and does not beat that benchmark. The latent-cohort lower bound is not a lower bound for learning the observable response vector; comparing it with FIR model-identification results for $\mathcal{H}_\infty$ norms would conflate different tasks

(Tu et al., 2018). No optimal estimation rate over all observation designs is claimed. The data cost is not an empirically measured exchange probing cost. An ordinary public-tape bridge, endogenous quotes and depth, incomplete resets, unknown or misspecified noise, and independent replication remain open. The information purchase in the example is predeclared. Its positive value is pointwise at the displayed economy; no prior over unknown economies or globally optimal Bayesian/robust acquisition policy is claimed. The fixed horizon, independent measurement contract, and conditional mean-zero future flows cannot be removed by the simulation. At another horizon the certificate must be recomputed. A rolling sequence needs simultaneous or time-uniform coverage; a one-decision guarantee is insufficient.

Literature priority. The reviewed sources already establish the general principles used here. The explicit dollar-cost/capital formulation and the combined result above are candidate additions to this project, not verified first-in-literature claims. A specialist must check whether they are an immediate corollary of existing task-specific or constrained-decision results before publication.

References

H. J. van Waarde, J. Eising, H. L. Trentelman and M. K. Camlibel. Data Informativity: A New Perspective on Data-Driven Analysis and Control. *IEEE TAC* 65, 4753–4768.

A. J. Wagenmaker, M. Simchowitz and K. Jamieson. Task-Optimal Exploration in Linear Dynamical Systems. *ICML*, PMLR 139, 10641–10652.

D. L. Donoho. Statistical Estimation and Optimal Recovery. *Annals of Statistics* 22, 238–270.

P. Heinrich and J. Kahn. Strong Identifiability and Optimal Minimax Rates for Finite Mixture Estimation. *Annals of Statistics* 46, 2844–2870.

D. Blackwell. Equivalent Comparisons of Experiments. *Annals of Mathematical Statistics* 24, 265–272.

S. Tu, R. Boczar and B. Recht. Minimax Lower Bounds for $\mathcal{H}_\infty$-Norm Estimation. *arXiv:1809.10855*.

Checked synthetic financial illustration

This illustration uses the frozen primary design and the stronger comparison recorded in the design amendment. It is exploratory and was developed after the earlier project runs; it is not independent preregistration or replication. The two-cohort primary has $a = 0.4$, $\kappa = 1$, $Q = 0.4$, rates $0.49, 0.51$, and equal weights. The known class cap is $\overline{Q} = 0.55$; $H = 60$ one-minute model periods, $N = 32$, $\sigma = 0.02$, and $\gamma = 0.025$. Initial wealth is 1,000, price 100, stress budget 1, and flow bound 0.04. The service fee is 0.08 per unit. Execution costs are $0.002I + 0.0002I^2$ and carry is charged at 5 percent annualized. The assumed all-in reset fee is 0.002, so the learning bill is 0.064. These are hypothetical model dollars, not an exchange calibration or returns. The same-error-budget no-data lottery is a theoretical strongest comparator, not a proposed desk practice. The zero-violation comparisons are also shown.

Policy	Mean units	Net expected profit	Unsafe reserve (%)
No data, zero violations	1.306	0.1007	0.00
Best no data, same error budget	1.523	0.1171	2.50
Finite-horizon certificate	4.553	0.2844	2.44
Standard Gaussian projection	4.553	0.2844	2.44
Full-response ellipsoid	4.386	0.2718	0.00
Point estimate (uncertified)	4.601	0.2880	50.41
Exact gain report, fee 0.05	4.669	0.3072	0.00
Response oracle, zero violations	4.914	0.3756	0.00
Response oracle, same error budget	5.041	0.3851	2.50

Profit integrates the future zero-mean flow model and every declared cost; the data-dependent policy values are also integrated over reading noise by one-dimensional Gaussian quadrature. Mean units and empirical reserve violations use 20,000 synthetic reading draws. Oracle and no-data lottery values are analytic. A reserve violation means some bounded future path can exceed the budget, not that a sampled market path actually did so. The 97.5 percent safety statement is over measurement draws for a single future commitment; it is not certainty, market-wide stability, or time-uniform coverage. The point rule buys slightly more expected profit by violating the same risk requirement and is not a valid superiority comparison. The proposed estimator and standard Gaussian projection coincide exactly.

The net information advantage over the strongest same-error-budget no-data rule is 0.1673. The written theorem's sufficient lower bound is 0.1465, strictly positive even after all costs. At the limiting merged economy the lower bound is 0.1465; continuity therefore supports the small-separation corollary. These bounds are evaluated in floating point with a wide positive margin, not claimed interval-arithmetic certificates.

Separation δ	Known-pair resets required	Implied fee bill
0.0800	478	0.955
0.0400	3.96e+04	79.2
0.0200	2.84e+06	5.67e+03
0.0100	1.91e+08	3.83e+05
0.0050	1.26e+10	2.51e+07
0.0025	8.15e+11	1.63e+09

The known-pair test receives all 60 taps, exactly as the financial rule does. Response differences are calculated in rational arithmetic before conversion to floating point. The sample threshold is the Gaussian testing formula, not a measured runtime or an asserted physical trading-capital requirement. For the primary $\delta = 0.01$, distinguishing the two already-known alternatives requires about 191 million resets at 5 percent error. Even this optimistic information fee is far beyond the unit stress budget. Identifying arbitrary cohorts cannot be easier than this embedded test under the same fixed query contract. Other experimental designs are not covered.

At fixed $N = 32$, the expected net benefit crosses zero at an illustrative reset price of about 0.005825. An exact scalar gain report is worth purchasing only below a fee of about 0.188690 in this example. These are computed break-even prices, not observed reporting costs.

Failures retained. The full output includes all 819 cost, horizon and service-fee sensitivity rows, including negative values and unaffordable information budgets. A low service premium or expensive measurements can make abstaining from data acquisition preferable. Independent checks cover 13 algebra, optimizer, shock-corner, cash-flow and Gaussian-testing properties, including the same-error-budget no-data optimum checked against a separate linear program. An initial low-premium

LP check was affected by the solver's absolute objective tolerance; rescaling the objective resolved it, and that failed log is retained. This is not independent mathematical proof review.

Reading/model contract	Gain coverage (%)	Unsafe reserve (%)
Independent Gaussian, valid	97.36	2.63
Correlated taps, wrong variance	71.16	28.84
Correlated taps, correct variance	97.36	2.63
Negative reading bias, invalid	0.00	100.00
Future feedback changes, invalid	0.00	100.00

The valid Gaussian row has finite Monte Carlo variation around its nominal 97.5 percent coverage. Using the correct correlated variance restores the specified measurement contract at the cost of a larger margin. Bias or future parameter change is not repaired by a larger sample under the wrong model. A signed kernel such as $(1, -0.9)$ has sum 0.1 and absolute sum 1.9, so positivity is indispensable to using the plain sum. The same certificate used repeatedly without correction does not retain its one-decision guarantee.

What changed economically. In this model, the desk can choose a larger inventory commitment while paying for information and retaining the same permitted unsafe-action probability as its strongest no-data comparator. This is an example of positive decision value, not evidence that information always pays, that trading strategies must be identified, or that a real venue's liquidity and welfare improve. Realistic observation costs and a public-data bridge remain requirements for an empirical finance claim.

